Number System

Decimal number System	Binary number System	octal number System	Hexadecimal number system
(10) (0-9)	(2)	(8) (0-7)	(0-9 and A-F)

1] Decimal number system :-

i] uses 10 well known symbols and their combination to form a new number is called as decimal number system

ii] The 10 well known symbols are : 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

iii] The base or radix of this system is 10

iv] The weight assign to these number system for integer parts are $10^0, 10^1, 10^2, 10^3 \cdots$ from LSB (least significant bit) and $10^{-1}, 10^{-2} \cdots$ are for fractional part from MSB bit (most significant bit)

v] for example :- $(235)_{10}$, $(136.51)_{10}$

235 — MSB, LSB

136.51 — MSB, LSB (integer part); 51 — MSB, LSB (fractional part)

$10^2 \leftarrow 136.51 \rightarrow 10^{-2}$

$10^1 \quad 10^0 \quad 10^{-1}$

2] Binary number system :-

i] uses 2 well known symbols and their combination to form a new number is called as Binary system

ii] The 2 well known symbols are : 0, 1

iii] The base or radix of this system is 2

iv] The weight assigned to this number system for integer parts are to $2^0, 2^1, 2^2 \cdots$ for LSB integral parts from LSB and $2^{-1}, 2^{-2}, 2^{-3} \cdots$ for fractional part from MSB

v] for example :- $(11100)_2$, $(101011)_2$

3] Octal number system :-

i] uses 8 well known symbols and their combinations to form a new number is called as octal number system

ii] The base or radix of this system is 8

iii] The 8 well known symbols are : 0, 1, 2, 3, 4, 5, 6, 7

iv] The weight assigned to this number system for integral parts are $8^0, 8^1, 8^2 \cdots$ from LSB and $8^{-1}, 8^{-2}, 8^{-3}$ for fractional part from MSB

v] for example :- $(247)_8$, $(153)_8$

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iii] 10 output devices :-

1] Plotter :- Mechanical pen devices drawing high resolution vector graphics

2] 3D Printer :- Gantry System synthesizing 3D physical objects

3] Braile Embosser :- Impact device pressing tactile dots onto paper

4] Speech Synthesizer :- Processor convert text into vocal audio

5] LCD Projector :- Lens system beaming video onto external surface

6] Microfilm output (COM) :- Camera transfering data onto microfilm

7] Haptic feedback device :- Acutator providing vibration sensation

8] Refreshable braile display :- Electro-mechanical pins raising to show text

9] GPS Navigation Screen :- dedicated display rendering real time positioning maps

10] Piezoelectric Buzzer :- Audio transducer generating specific warning beep frequencies

iii] Short/concise "History of Computers" :-

Charles Babbage is the "Father of the Computer", for designing the Analytical Engine in the 1830s, — the first conceptual general purpose mechanical computer

Later in 1945 John Von Neumann introduced the "von Neumann Architecture" which established the modern system design of using CPU, memory and shared data storage

The hardware history then evolved through five distinct generations, starting with vaccum tubes in 1940s and transitioning to transistors in the 1950s. The invention of integrated circuits (ICs) in the 1960s further shrank the system sizes, leading to the microprocessors of the 1970s that power our modern laptops and microphones

Today, the fifth generation focuses on Artificial Intelligence and parallel processing, continuing the shift from massive, room sized machines to highly integrated personal devices

iv] Concise note on "Current Generation Computers"

The fifth generation uses ultra Large Scale Integration (ULSI) tech to pack millions of components onto one tiny microchip. Instead of using one brain, these computers use parallel processing to run multiple brains at same time

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4] Hexadecimal number System :-

i] Uses 16 well known symbols and their combination to form a new number is called as Hexadecimal number system.

ii] The 16 well known symbols are : 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
A, B, C, D, E, F

iii] The base or radix of this system is 16

iv] The weight assigned to this system for integral parts are from $16^0, 16^1, 16^2 \cdots$ for LSB and $16^{-1}, 16^{-2}, 16^{-3} \cdots$ are for fractional part from MSB

v] for example :- $(C416E)_{16}$, $(BAD1)_{16}$

vi] Application :- colours in html (eg :- # FFFFFF)

HW Ans :-] 10 Input devices

1] Barcode reader
2] Biometric Scanner
3] Touch Pad
4] Magnetic Ink character Reader (MICR) -> Scanner reads magnetized ink
5] Optical Mark Reader (OMR) -> Scanner detects pencils mark
6] Graphic Tablet -> Digitize stylus movements into co-ordinate data
7] Magnetic Stripe reader -> Sensor read data from card magnetic bands
8] RFID Reader -> receiver gathering data from radio frequency tags
9] Light Pen -> Light sensitive wand pointing directly at CRT screens
10] MIDI keyboard -> controller sending musical notes data to Computer

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" Conversion b/w number system :-

1) Decimal to Binary (Double Dabble method) :

i) Divide decimal by 2
ii) Record remainder
iii) Divide quotient again by 2
iv) Repeat the process until equotient is 0
v) Read the remainder in reverse order (bottom to top)

eg:- $(34)_{10} = (?)_2$

2	34	
2	17	→ 0
2	8	→ 1
2	4	→ 0
2	2	→ 0
2	1	→ 0
	0	→ 1

quotient ↑ remainder

eg:- $(15)_{10} = (?)_2$

2	15	
2	7	→ 1
2	3	→ 1
2	1	→ 1
	0	→ 1

$(15)_{10} = (1111)_2$

$(34)_{10} = (100010)_2$

Decimal to binary (fractional part) :

15.125

integer :-

2	15	
2	7	→ 1
2	3	→ 1
2	1	→ 1
	0	→ 1

$(15)_{10} = (1111)_2$

fractional :- Fraction × Base = Recorded integer Part

$0.125 \times 2 = 0.250 \rightarrow 0$

$0.250 \times 2 = 0.500 \rightarrow 0$

$0.500 \times 2 = 1.000 \rightarrow 1$

↳ till fraction is 0

$(0.125)_{10} = (0.001)_2$

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Fractional part :- Fraction x Base = Result Result integer

$(786.55)_{10} = (?)_8$

8	786	0.55 x 8 = 4.40	4
8	98 → 2	0.40 x 8 = 3.20	3
8	12 → 2	0.20 x 8 = 1.60	1
8	1 → 4	0.60 x 8 = 4.80	4
	0 → 1	0.80 x 8 = 6.40	6

$(786)_{10} = (1422)_8$

- In the fraction part if the decimals are not becoming zero after multiple steps the stop when decimal starts repeating

c.w:- $(379.05)_{10} = (?)_8$

8	379	0.05 x 8 = 0.40	0
8	47 → 3	0.40 x 8 = 3.20	3
8	5 → 7	0.20 x 8 = 1.60	1
	0 → 5	0.60 x 8 = 4.80	4
		0.80 x 8 = 6.40	6

$(379)_{10} = (573)_8$

$(0.05)_{10} = (0.03146)_8$

$\therefore (379.05)_{10} = (573.03146)_8$

c.w:- $(379.05)_{10} = (?)_8$

8 | 379 — 0.05 x 8 = 0.40 — 4

0.05

0.05 x 8 = 0.4 → 0

0.4 x 8 = 3.2 → 3

$(0.05)_{10} = (0.03)_8$

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eg:- 1 1 . 1 0 1

2^1 2^0 2^{-1} 2^{-2} 2^{-3}

$(1\times2^1)+(1\times2^0)+(1\times2^{-1})+(0\times2^{-2})+(1\times2^{-3})$

$= 2 + 1 + 0.5 + 0 + 0.125$

$= 3.625$

$\therefore (11.101)_2 = (3.625)_{10}$

C.W:- $(111.110)_2$; 1 1 1 . 1 1 0

2^2 2^1 2^0 2^{-1} 2^{-2} 2^{-3}

$= (1\times4)+(1\times2)+(1\times1)+(1\times0.5)+(1\times0.25)+(0\times0.125)$

$= 4+2+1+0.5+0.25+0$

$= 7 + 0.75$

$= 7.75$

$\therefore (111.110)_2 = (7.75)_{10}$

5] Binary to octal number

- For this number system, base for binary is 2 base for octal no. system is 8.
- The base for octal no. is the 3rd power of the base for binary number
- Group 3 bits of binary number, starting from LSB for integer part MSB for fractional part, add zero (es) at end if required
- Write equivalent octal number for each group of 3 bits

eg:- $(10100110.1101)_2$

010 100 110 110 100 → jitne digit hai utna multiple of 2 likho (1, 2, 4, 8, 16 ⋯) then compare :-

2 4 6 6 4

246.64

(010 (421) 0 2 0) agar zero aya hoga to 0 hi

3] Decimal to hexadecimal

Steps :- Divide decimal by 16
- Record remainder
- Divide quotient again by 16
- Repeat the process until quotient is 0
- Read the remainder in reverse order

eg :- $(2121)_{10} = (?)_{16}$

16	2121	
16	132	9
16	8	4
	0	8

$(2121)_{10} = (849)_{16}$

C.W :- $(478)_{10} = (?)_{16}$

16	478	
16	29 → 14	
16	1 → 13	
	0 → 1	

$(478)_{10} = (1DE)_{16}$

A = 10
B = 11
C = 12
D = 13
E = 14
F = 15

4] Binary to decimal

- write a binary number.
- Assign corresponding Binary weights to each Binary Bit from LSB (2^0) to MSB (2^{n}) for integer
  from MSB (2^{-1}) to LSB (2^{-n}) for fractional
- multiply each Binary bit with corresponding weight
- Add all multiplications to the result

$2^0 = (1 \times 1) + (2 \times 1) + (1 \times 1) + (5 \times 1) + (2 \times 1)$

$2^{-1} = \frac{1}{2^1} = \frac{1}{2} = 0.5$

$2^{-2} = \frac{1}{2^2} = \frac{1}{4} = 0.25$

$2^{-3} = \frac{1}{2^3} = 0.125$

$2^{-4} = \frac{1}{2^4} = 0.0625$

$2^{-5} = \frac{1}{2^5} = 0.03125$

fractional → easy calculation

integer

$2^0 :- 2^0 \times 1 = 1$

$2^1 :- 2^1 \times 1 = 2$

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8] Octal to Binary number system

1] $(147.750)_8 = (?)_2$

? $(147.750)_8$ = $(001100111.111101000)_2$

1	4	7	.	7	5	0
↓	↓	↓		↓	↓	↓
001	100	111	.	111	101	000

2] $(345.012)_8 = (?)_2$

3	4	5	.	0	1	2
↓	↓	↓		↓	↓	↓
011	100	101		000	001	010

$(345.012)_8 = (011100101.000001010)_2$

9] Octal to Hexadecimal number system

$(4567.666)_8 = (?)_{10}$

? $(4567.666)_8$ = $(977.DB0)_{16}$

4	5	6	7	.	6	6	6
↓	↓	↓	↓		↓	↓	↓
100	101	110	111		110	110	110

100101110111.110110110

1001	0111	0111	.	1101	1011	0000
↓	↓	↓		↓	↓	↓
9	7	7		13	11	0

$(6130.125)_8$

6	1	3	0	.	1	2	5
110	001	011	000		001	010	101

110001011000.001010101×××

1100 0101 1000 . 0010 1010 1000

12	5	8		2	A	8

$\therefore (6130.125)_8 = (C58.2A8)_{16}$

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6) Binary to Hexadecimal number conversion

$(10111011.10101111)_2 = (?)_{16}$

0001	0111	0111 .	1010	1111	1000
↓	↓	↓	↓	↓	↓
1	7	7	10	15	8
↓	↓	↓	↓	↓	↓
1	7	7	A	F	8

$(10111011.10101111)_2 = (177.AF8)_{16}$

$(11011.101)_2 = (?)_{16}$

11011.101

0001	1011 .	1010
↓	↓	↓
1	11	10
↓	↓	↓
1	B	A

$(11011.101)_2 = (1B.A)_{16}$

7) Octal to Decimal number system

$(345)_8 = (?)_{10}$

3	4	5
↓	↓	↓
8^2	8^1	8^0
↓	↓	↓
3×8^2	4×8^1	5×8^0

$= 3(64) + 32 + 5$

$= 192 + 32 + 5$

$= 229$

∴ $(345)_8 = (229)_{10}$

$(441.69)_8$

4	4	1 .	6	9
↓	↓	↓	↓	↓
8^2	8^1	8^0	8^{-1}	8^{-2}

$\therefore 4(8^2) + 4(8^1) + 1(8^0) + 6/8 + 9/64$

$256 + 32 + 1 + 0.75 + 0.14$

$= 289.89$

$\therefore (441.69)_8 = (289.89)_{10}$

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Binary Arithmetic

1] Binary Addition rule

$n = 2 \quad 2^n = 2^2 = 4$

A	B	Sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$n = 2 \quad (1,0)$

$2^n = 2^2 = 4$

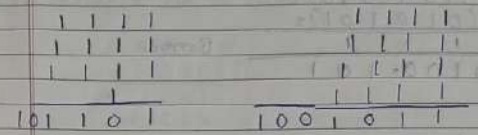
$(9)_{10} + (15)_{10} = (?)_2$

$(9)_{10} = (1001)_2$

$(15)_{10} = (1111)_2$

```
  1 0 0 1  -> (9)10
+ 1 1 1 1  -> (15)10
-----------
1 1 0 0 0  -> (24)10
```

$\therefore (24)_2$



2] Binary subtraction rule

A	B	Diff	borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Q $(3)_{10} - (2)_{10} = (?)_2$

$$\begin{array}{r} (0011)_2 \\ -(0010)_2 \\ \hline 0001 \end{array}$$

$= (0001)_2 = (1)_{10}$

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10] Hexadecimal to Decimal number system

$(BAD.1C)_{16} = (?)_{10}$

B	A	D	.	1	C
↓	↓	↓		↓	↓
16^2	16^1	16^0		16^{-1}	16^{-2}

$= B \times 16^2 + A(16) + D(16)^0 + 1(16)^{-1} + C$

$= 11(256) + 10(16) + D(1) + 1/16$

$= 2816 + 160 + 13 + 0.0625$

$= (2989.0625)_{10}$

11] Hexadecimal to Binary number system

$(DFCB1 \cdot BAE)_{16}$

D	F	C	B	1	.	B	A	E
↓	↓	↓	↓	↓		↓	↓	↓
1101	1111	1100	1011	0001		1011	1010	1110

$(1101111111001011000110111010111110)_{B2}$

12] Hexadecimal to octal number system

$(142.ABCD)_{16}$

1	4	2	.	A	B	C	D
↓	↓	↓		↓	↓	↓	↓
0001	0100	0010		1010	1011	1100	1101xx
3 digit	3 digit	3 digit		3	3	3	3 [0 if x, no digit]

000	101	000	010	.	101	010	111	100	110	100
↓	↓	↓	↓		↓	↓	↓	↓	↓	↓
0	5	0	2		5	2	7	4	6	4

$(0502.527464)_8$

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" Binary Division

Divisor ← 3) 7 ← Dividend
2 → Quotient
- 6
1 → remainder

Dividend = Divisor × Quotient + remainder

```
      10
11 ) 111
   - 11↓
     001
   - 000
     1
```

∴ $(10)_2 = (2)_{10}$

Q $(25)_{10} \div (5)_{10}$

$(11001)_2 \div (101)_2$

```
      110
101 ) 11001
    - 101||
      1 ↓↓ Borrow
      0010|
    -  101
       0
```

```
      101
101 ) 11001
    - 101
      00101
    -  101
       0
```

$(11)_2$ = $(101)_2 = (5)_{10}$

Q $(25)_{10} \div (5)_{10}$

$(11001)_2 \div (101)_2$

```
      101
101 ) 11001
    - 101
      1
      00101
    -  101
       0
```

$(101)_2 = (5)_{10}$

" octal arithmetic :-

1] octal Addition :-

Rule :- Add the colmun normally if a sum is 8 or more subtract 8 and carry 1 to the next colmun.

eg :-

```
    76
    24
  +
    1      carry
   10 10   ..... 8 or 8+ ∴ 8 subtract kiya aur 1 carry to next colmun
  - 8  8
  (12  2)8   final
```

Partial sum

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Q 5531
3261

```
  101 101 011 001
- 011 010 110 001
  ---------------
  010 010 101 000
    2   2   5   0
```

$(2250)_8$

" Negative no. representation in Binary system :-

For representing signed negative no. we have to follow following method :-

1] Sign magnitude code (SM Representation)

2] One's complement method (1's complement)

3] Two's complement method (2's complement)

1] SM Representation :-

Format :- Number B = $\underbrace{B_{n-1}}_{1}$ $\underbrace{B_{n-2}\ B_{n-3} \cdots B_2\ B_1\ B_0}_{n-1}$

MSB represent Sign Bit magnitude (IBI) Bit.
Sign bit

normally sign bit (S bit) = 0 indicates positive number and (S bit) = 1 indicates negative number

Q Convert $(124)_{10}$ into binary
$(-124)_{10}$

$(125)_{10} = (1111100)$

+124 :- 0 1 1 1 1 1 0 0
B_7 B_6 B_5 B_4 B_3 B_2 B_1 B_0

-124 :- 1 1 1 1 1 0 0 0
B_7 B_6 B_5 B_4 B_3 B_2 B_1 B_0

Jas Num ne age 0 use kiya ∴ + value age 1 use kiya ∴ - value

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2] $1^{'s}$ Complement :-

In this method zero is replaced with 1 and one is replaced with 0

Q $1^{'s}$ Complement of 9

$(9)_{10} = (1001)_2$

$\therefore$ $1^{'s}$ Complement $= (0110) = (-9)_{10}$

3] $2^{'s}$ Complement :-

In this method we will add 1 to $1^{'s}$ Complement of the number

formula :- $2^{'s}$ Complement $= 1^{'s}C + 1$

Q $2^{'s}$ C of $(9)_{10}$

$(9)_{10} = (1001)_2$

$1^{'s} = 0110$

$2^{'s} = 0110 + 1$

$= 0111$

$\therefore 2^{'s} = (0111)_2$

" $1^{'s}$C subtraction method :

Step 1 :- write binary of A and B for A-B

Step 2 :- write $1^{'s}$C of B

Step 3 :- Add step 1 and 2

Step 4 :- if "carry" is generated perform

EAC [Endaround carry] i.e add carry to the result and result is in true form and Positive

Otherwise result is negative and it is in $1^{'s}$C form

" $2^{'s}$ Complement subtraction method :

Step 1 :- write binary of A and B for A-B

Step 2 :- write $2^{'s}$ C of B

Step 3 :- Add step 1 and 2

Step 4 :- if "carry" is generated perform :

Discard the carry and the result is in true form and Positive

Otherwise result is negative and it is in $2^{'s}$C form

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Q. 624 − 154 = 888 − 888 = 1000

$= (1000)_8$

Q. 523 − 167 = 910 − 88 = 712

$= (712)_8$

Q. $(727)_{10} = (1163)_8$

727 / 8	90 + 7
90 / 8	11 + 2
11 / 8	1 + 3
1 / 8	0 + 1

Reading the remainders from bottom to top: 1163

2) Octal Subtraction :

- Direct method :

if minuend is less than subtractedend we borrow 8 and carry 1

eg :- 5 → minuend
−4 → subtrahend
1 → Difference

Q. 751 − 273 = 456 (8 borrow, carry)

Q. 653 − 77 = 554 (8 + borrow, carry)

Q. 5531 − 3261 = 2250 (8 + borrow, carry)

- Conversion method

Q. 751 − 273

751 → 111 101 001
273 → 010 111 011
1's complement → 101 000 100
1 → 1 111 1
100 101 110

$(4\ 5\ 6)_8$

Q. 653 − 77

653 → 110 101 011
77 → 111 111
1's complement → 000 111 100
11 111 100
101 101 100

$(554)_8$

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Q Perform 8-9, using 1^{s}C and 2^{s}C Subtraction method.

1] 1^{s} complement Subtraction method :-

$(8)_{10} = (1000)_2$

$(9)_{10} = (1001)_2$

$1^{s}C\ (9)_{10} = (0110)_2 = -9$

$$\begin{array}{rl} 8 & = 1000 \\ +(-9) & + 0110 \\ \hline \square & 1110 \\ \downarrow & \end{array}$$

no carry generated

∵ carry is not generated

∴ result will be negative

∴ 1^{s}C of result = $0001 = (-1)_{10}$

2] 2^{s} complement Subtraction method

$(8)_{10} = (1000)_2$

$(9)_{10} = (1001)_2$

$1^{s}C\ (9)_{10} = (0110)_2$

$2^{s}C\ (9)_{10} = 1^{s}C + 1 = 0110 + 1$

$= 0111$

$$\begin{array}{rl} 8 & = 1000 \\ +(-9) & \quad 0111 \\ \hline \square & 1111 \\ \downarrow & \end{array}$$

∵ no carry is generated

∴ result will be negative

∴ 2^{s}C of result = 1^{s}C of result + 1

$= 0000 + 1$

$= 0001$

$= (-1)$

↳ negative cause no carry.

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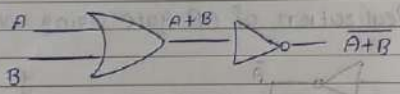
Realization of NOR gate using NAND gate

Implementation using NAND gate :

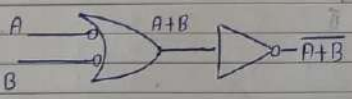
- For the given logic function, draw the logic circuit using the basic gates (AND, OR, NOT gate)
- Add bubble at the input side of OR gate and at the output side of AND gate
- Place a NOT gate wherever bubbles are added in previous step
- Eliminate any double inversion
- Replace all gates with NAND gate

The boolean eqⁿ of NOR gate = $\overline{A+B}$

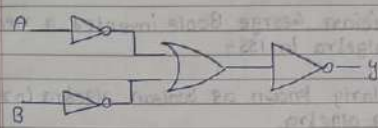
Step 1 :



Step 2 : Input ~~at~~ of OR gate with bubble

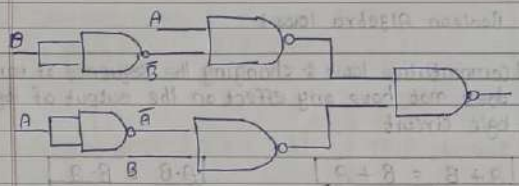


Step 3 : Place Not gate at bubble



Step 4 : No double inversion here

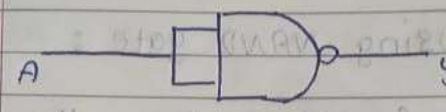
Step 5 : Replacing all gates with NAND



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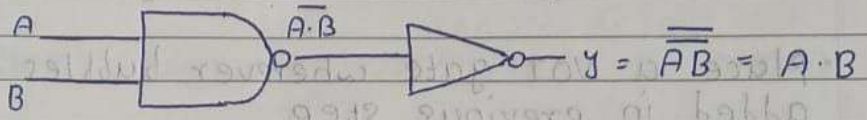
Logic Gates

" Realization of NOT gate using NAND gate.

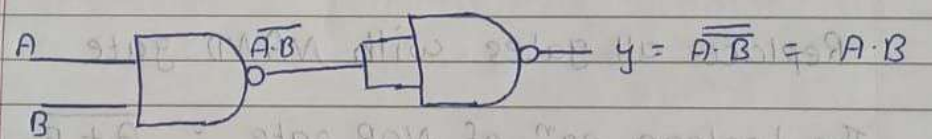


since $A \cdot A = A$

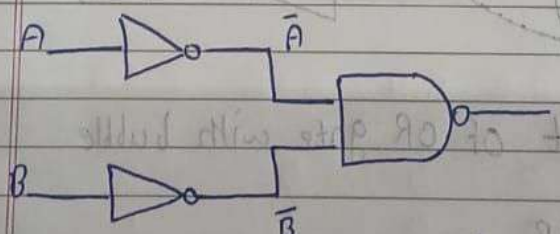
" Realization of AND gate using NAND gate



or



" Realization of OR gate using NAND gate



$y = A + B$
$= \bar{A} + \bar{B}$
$= \overline{A \cdot B} = A + B$

OR gate

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Boolean Algebra :-

- Mathematician George Boole invented a new kind of algebra in 1854
- It is popularly known as boolean algebra (or) switching algebra.
- Boolean algebra is used to analyse and simply the digital circuits.

Boolean Algebra laws :-

1] Commutative laws :- changing the sequence of variable does not have any effect on the output of logical logic circuit

$A + B = B + A$ — Commutative law of addition (or)

$A \cdot B = B \cdot A$ — Commutative law of multiplication (and)

Proof :-

Input A	Input B	A·B	B·A	A+B	B+A
0	0	0	0	0	0
0	1	0	0	1	1
1	0	0	0	1	1
1	1	1	1	1	1

2] Associative law :- order in which the logic operation are performed is irrelevant as their effect is same

(a) $(A \cdot B)C = A \cdot (B \cdot C)$ — Associative law of multiplication

(b) $(A + B) + C = A + (B + C)$ — Associative law of addition

3] Distributive law :- $A \cdot (B + C) = A \cdot B + A \cdot C$

4] Idempotent law :- Theorem 1 :-

i) $A + A = A$ ii) $A \cdot A = A$

Theorem 2 :- $A + 1 = 1$

Theorem 3 :- $A \cdot 0 = 0$

} null element

5] Absorption law :- Theorem 4 :-

(i) $A + AB = A$

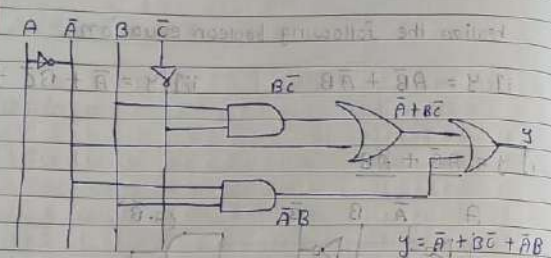
(ii) $A(A + B) = A$

(iii) $A(\bar{A} + B) = AB$

(iv) $(A + B)(A + C) = A + BC$

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2] $y = \bar{A} + B\bar{C} + \bar{A}B$



* Standard representation of logic function :-

The logic functions are represented in terms of logical variable. The value of these variables are in the form of binary bits 1 and 0. Any logical function can be expressed in following form

(i) SOP (Sum of Product)
(ii) Se POS (Product of sum)

* Combination circuit is circuit whose o/p depends on current I/p

1] Sum of product :- logical function is in the form of sums of different products
like $y = ABC + A\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$

Using sum of products form to realize the circuit always use NAND Gate

2] Product of sum :- logical function is in the form of product different sums
like $y = (A+B+C)(A+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})$

Using POS form to realize the circuit always use NOR gate.

* Maxterm and Minterm :- ? used in k map

(POS) use hoga	(SOP) use hoga
variable 0 se	variable 1 se
variable bar 1 se	variable bar 0 se
represent	represent
↓	↓
variable = 0	variable = 1
variable bar = 1	variable bar = 0

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